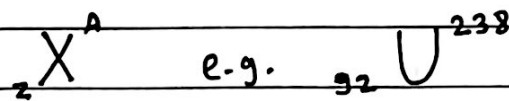


Nuclear Physics

- ↳ Nuclear physics is the study of nucleus of an atom.
- ↳ Nucleus consists of proton and neutron which are collectively known as nucleons.
- ↳ A nucleus is represented by



Where, $Z = \text{Atomic number}$
 $A = \text{Atomic mass}$

• Atomic number

- ↳ The Number of protons of an atom is called Atomic number.
i.e. $Z = \text{number of protons}$.

• Mass Number (A)

- ↳ The sum of number of protons and number of neutrons present in the nucleus is called mass number.

$$\begin{aligned} \therefore \text{Mass Number} &= \text{Number of Proton} + \text{Number of neutrons} \\ &= \text{Sum of Number of Nucleons} \\ &\quad \text{of a nucleus.} \end{aligned}$$

• Properties of Nucleus

① ↳ Nuclear size

- The Nuclear size depends upon mass

number. The empirical relation for the nuclear radius is given by

$$R = r_0 A^{1/3}$$

Where $r_0 = 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fm}$

(i) Nuclear charge

- It is the charge of nucleus. A nucleus consists of proton and neutrons. Proton are positively charged particle and neutron is chargeless particle. So, Nucleus carry positive charge which is given by

$$Q = Z + Ze \quad \text{Where, } e = 1.6 \times 10^{-19} \text{ C is the electronic charge}$$

(ii) Nuclear Mass

- It is the sum of masses of protons and neutrons present in the nucleus.

$$\therefore \text{Nuclear Mass (M)} = Zm_p + (A - Z)m_n$$

Where, $m_p \rightarrow$ mass of a proton
 $m_n \rightarrow$ mass of a neutron.

Since, $m_p \approx m_n$

$$\begin{aligned} \therefore M &= Zm_p + (A - Z)m_n \\ &= M_n (Z + A - Z) \\ &= AM_n \end{aligned}$$

✓ (iv) Nuclear density

- The mass per unit volume of nucleus is called Nuclear density.

$\therefore$ Nuclear density (ρ_n) = $\frac{\text{Nuclear mass}}{\text{Volume of nucleus}}$

$$= \frac{A m_n}{\frac{4}{3} \pi R^3}$$

$$= \frac{A M_n}{\frac{4}{3} \pi (r_0 A^{1/3})^3}$$

$$= \frac{\cancel{A} M_n}{\frac{4}{3} \pi r_0^3 \cancel{A}}$$

$$= \frac{3 M_n}{4 \pi r_0^3}$$

This shows that Nuclear density is independent of mass Number.

i.e. Nuclear density of all the Nuclei is almost same.

Since, $m_n = 1.66 \times 10^{-27} \text{ kg}$ and $r_0 = 1.2 \times 10^{-15} \text{ m}$
then,

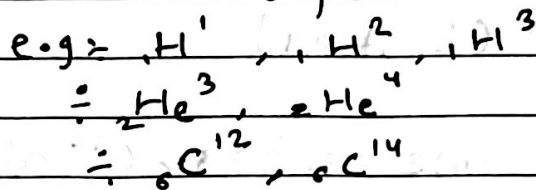
$$\begin{aligned} \rho_n &= \frac{3 \times 1.66 \times 10^{-27}}{4 \times 3.14 \times (1.2 \times 10^{-15})^3} \\ &\approx 2.3 \times 10^{17} \text{ kg/m}^3 \end{aligned}$$

This shows that the density of Nucleus is extremely high.

• Types of Nuclei

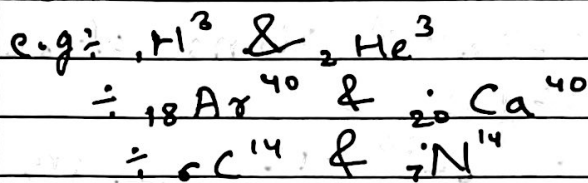
(i) Isotopes

→ The Nuclei having same Atomic number but different mass no are called isotopes.



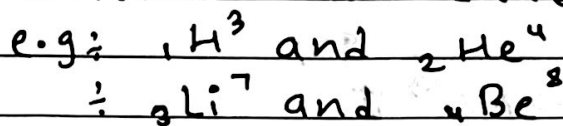
(ii) Isobars

→ The nuclei having same mass no but different atomic no are called Isobars.



(iii) Isotones

→ The Nuclei having same no of neutrons are called isotones.



• Einstein's mass energy relation

→ According to Einstein, Mass and Energy are interconvertible. The energy equivalent to mass Δm is given by

$$E = \Delta mc^2$$

Where, $c = 3 \times 10^8 \text{ m/s}$ is speed of light in vacuum.

This is known as Einstein's mass energy relation.

- Atomic mass unit (1 Amu)

$\hookrightarrow \left(\frac{1}{12}\right)^{\text{th}}$ of mass of one ~~carbon~~ ^{12}C atom

is called one Amu.

Simply, 1 Amu is the mass of 1 proton or neutron.

Since mass of 6.023×10^{23} atoms of ^{12}C = 12 gm

$\therefore$ mass of 1 atom of $^{12}\text{C} = \frac{12}{6.023 \times 10^{23}} \text{ gm}$

Then,

1 amu = $\left(\frac{1}{12}\right)^{\text{th}}$ of mass of one ^{12}C atom

$$= \frac{1}{12} \times \frac{12}{6.023 \times 10^{23}} \text{ gm}$$

$$= 1.67 \times 10^{-24} \text{ gm}$$

$$= 1.67 \times 10^{-27} \text{ Kg}$$

- Energy equivalent to 1 amu

$\hookrightarrow$ Here,

$$m = 1 \text{ amu}$$

$$= 1.67 \times 10^{-27} \text{ Kg}$$

Then,

$$\begin{aligned}
 E &= mc^2 \\
 &= 1.67 \times 10^{-27} \times (3 \times 10^8)^2 \text{ Joule} \\
 &= 1.494 \times 10^{-10} \text{ J} \\
 &= \frac{1.494 \times 10^{-10}}{1.6 \times 10^{-19}} \text{ eV} \quad [\because 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}] \\
 &\approx 931 \times 10^6 \text{ eV} \\
 &\approx 931 \text{ MeV}
 \end{aligned}$$

• Mass defect (Δm)

→ The difference in the sum of masses of Nucleons of nucleus and rest mass of nucleus is called mass defect.

$$\therefore \text{mass defect}(\Delta m) = Zm_p + (A-Z)m_n - M$$

Where $m_p \rightarrow$ mass of proton
 $m_n \rightarrow$ mass of a neutron
 $M \rightarrow$ rest mass of Nucleus.

• Packing fraction (f)

→ The mass defect per nucleon is called packing fraction.

$$\therefore \text{Packing fraction} = \frac{\Delta m}{A}$$

• Binding Energy

- The minimum energy required to bind the nucleons to form a stable

nucleus is called Binding energy.

or

↳ The minimum energy required to disintegrate (or separate) the nucleons from the nucleus is called binding energy.

Binding energy is the energy equivalent to mass defect.

$$\therefore B.E = \Delta mc^2$$

$$= [Zm_p + (A-Z)m_n - M]c^2$$

Note: If mass defect is in Amu then
Binding energy = $\Delta m \times 931 \text{ MeV}$

• Binding energy per ~~nucleon~~ Nucleon

↳ The minimum energy ~~separ~~ required to disintegrate ~~the~~ a nucleon from the nucleus is called binding energy per Nucleon.

$$\therefore \text{BE per nucleon } (\overline{B.E}) = \frac{B.E}{A}$$

* Binding energy per nucleon explains the stability of the nucleus.
↑ For the binding energy per nucleon,

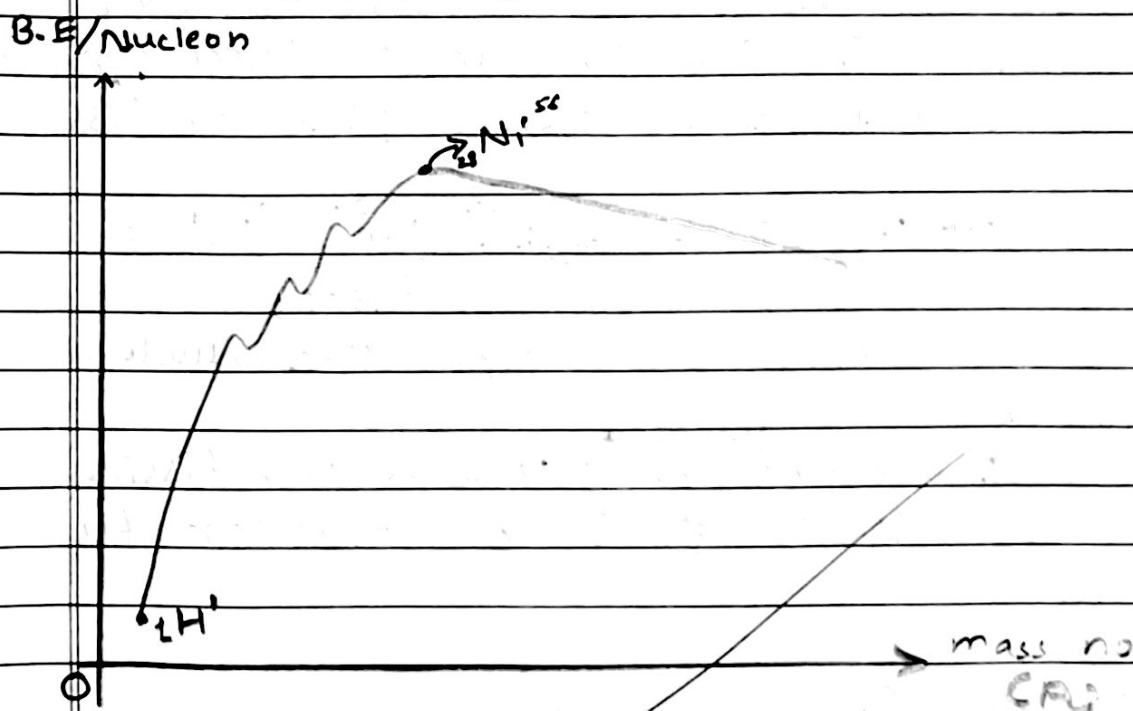
more is the stability of nucleons.
And vice versa.

• Significance of Binding energy per nucleon

↳ The binding energy per nucleon explains the stability of nucleons.

Higher the binding energy per nucleon, higher is the stability of nucleons and vice versa.

• Binding Energy Curve



> The binding energy per nucleon of ${}_{28}^{56}\text{Ni}$ is maximum. i.e. 8.795 MeV.
So, Nickel is most stable.

> The binding energy per nucleon of lighter nuclei is less so they are

less stable. For the stability, they undergo nuclear fusion.

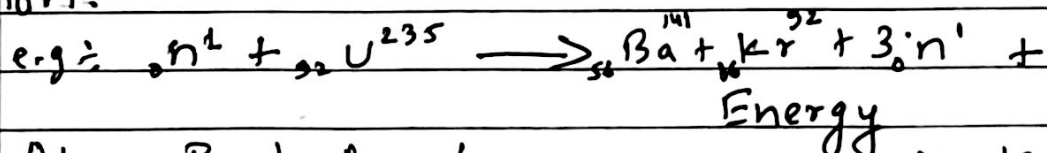
> The ~~be~~ binding energy per nucleon of heavy nuclei (e.g. ${}_{92}\text{U}^{235}$) is less.

So they are unstable.

For stability, They undergo nuclear fission.

• Nuclear fission

→ The process of splitting of a heavy nucleus into two or more lighter nuclei with the release of large amount of energy is called Nuclear fission.

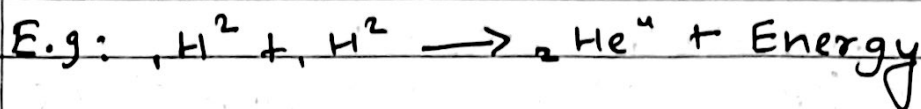


* Atom Bomb is based upon the Nuclear fission.

* ~~Binding~~ Energy^{released} per nucleon per fission is less than that of nuclear fusion.

• Nuclear fusion

- The process of combination of two or more lighter nuclei to form a heavy nucleus with the release of large amount of energy is called Nuclear fusion.



- * Hydrogen bomb is based upon the nuclear fusion.
- * Energy released per nucleon is more in nuclear fusion than that in nuclear fission.

Note: Neutron is used as bombarding particle in the nuclear fission because it is electrically neutral.

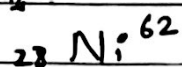
Homework

Numerical

- ① ${}_{28}\text{Ni}^{62}$ may be described as the most strongly bound nucleus because it has the highest B.E per nucleon. Its neutral at. mass is 61.928349 amu. Find its mass defect, its total binding energy and binding energy Nucleon.

Ans Soln

→



$M = 61.928349 \text{ amu}$

$\Delta m = ?$

$B.E = ?$

$B.E / \text{nucleon} = ?$

$m_n = 1.008665 \text{ amu}$

$m_p = 1.007825 \text{ amu}$

$1 \text{ amu} = 931.5 \text{ MeV}$

Now,

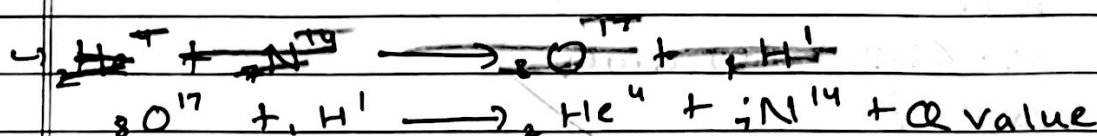
$$\begin{aligned}
 \textcircled{i} \quad \Delta m &= Z m_p + (A-Z) m_n - M \\
 &= 28 (1.007825) + (62-28) \times 1.008665 \\
 &\quad - 61.928349 \\
 &= 28.2191 + 34.29461 - 61.928349 \\
 &= 0.585361
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{ii} \quad B.E &= \Delta m \times 931.5 \text{ MeV} \\
 &= 0.585361 \times 931.5 \\
 &= 545.2637715
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{iii} \quad B.E / \text{nucleon} &= \frac{B.E}{A} \\
 &= \frac{545.2637715}{62} \\
 &= 8.79
 \end{aligned}$$

③ Calculate the Q-value of the reaction and mention the type of reaction.

$$\begin{aligned}
 {}_2\text{He}^4 &= 4.00377 \text{ amu} & {}_8\text{O}^{17} &= 17.00450 \text{ amu} \\
 {}_7\text{N}^{14} &= 14.00783 \text{ amu} & {}_1\text{H}^1 &= 1.00814 \text{ amu}
 \end{aligned}$$



$$\begin{aligned}
 \Delta m &= \text{mass of reactant} - \text{mass of product} \\
 &= (1.00814 + 17.00450) - (4.00377 + 14.00783) \\
 &= 1.04 \times 10^{-3}
 \end{aligned}$$

$$Q\text{-value} = \Delta m \times 931 \text{ MeV} \\ = 0.96824 \text{ MeV}$$

It is exothermic reaction because 0.96824 MeV is released.

✓/4) The energy liberated in the fission of a single uranium-235 atom is $3.2 \times 10^{-11} \text{ J}$. Calculate the power production corresponding to the fission of 1 kg of uranium per day.

→ Soln.

$$E \text{ per U-235 atom} = 3.2 \times 10^{-11} \text{ J}$$

$$\text{Power (P)} = ?$$

$$\text{mass (m)} = 1 \text{ kg} = 1000 \text{ g}$$

$$\text{time (t)} = 1 \text{ day} = 1 \times 24 \times 60 \times 60 \text{ sec} \\ = 86400 \text{ sec}$$

$$N_A = 6.0 \times 10^{23} \text{ mole}^{-1}$$

$$\text{Since, } \frac{N}{N_A} = \frac{m}{M}$$

$$\text{or, } \frac{N}{6.023 \times 10^{23}} = \frac{1000}{235}$$

$$\text{or, } N = \frac{1000 \times 6.023 \times 10^{23}}{235}$$

$$\therefore N = 2.55 \times 10^{24} \text{ atoms.}$$

Now,

$$\text{Total energy released (E)}$$

$$= 2.55 \times 10^{24} \times 3.2 \times 10^{-11}$$

$$= 8.16 \times 10^{13} \text{ J}$$

$$\begin{aligned}\text{Then } P &= \frac{E}{t} \\ &= \frac{8.16 \times 10^{13} \text{ watt}}{86400} \\ &= 9.4 \times 10^8 \text{ watt}\end{aligned}$$

② Estimate the binding energy per nucleon of ${}^7_3\text{Li}$. Mass of ${}^7_3\text{Li}$, a proton and a neutron are respectively 7.01435 amu, 1.00728 amu and 1.00867 amu.

→ Soln,

$$Z=3, A=7$$

$$M_p = 1.00728$$

$$M_n = 1.00867$$

$$M = 7.01435$$

$$\begin{aligned}\Delta m &= Zm_p + (A-Z)m_n - M \\ &= 3 \times 1.00728 + (7-3)1.00867 - 7.01435 \\ &= 3.02184 + 4.03468 - 7.01435 \\ &= 0.04217\end{aligned}$$

Now,

$$\begin{aligned}\text{B.E} &= \Delta m \times 931 \\ &= 0.04217 \times 931 \\ &= 39.26027\end{aligned}$$

Last,

$$\begin{aligned}\text{B.E/nucleon} &= \frac{\text{B.E}}{A} \\ &= \frac{39.26027}{7} \\ &= 5.6 \text{ Mev/nucleons}\end{aligned}$$

• Discussion Questions.

1) The nuclear density is almost constant for all nuclei. Why?

Ans. Nuclear density = $\frac{\text{Mass of nucleus}}{\text{Volume of nucleus}}$

The volume of the nucleus is directly proportional to the mass number A (i.e. to the no. of nucleons). If the numbers of nucleons are doubled, volume is also doubled. The nuclear density is independent of mass number. As both the mass and volume of the nucleus are increased by the same ratio on increasing the number of nucleons, hence all nuclei have nearly the same density. Its value is about $2.3 \times 10^{17} \text{ kg m}^{-3}$.

2) Define atomic mass unit and show that $1 \text{ amu} = 931 \text{ MeV}$.

Ans. Atomic mass unit is defined as exactly $\frac{1}{12}$ of the mass of a single neutral atom of carbon-12 in its ground state.

Here,

$$1 \text{ amu} = \frac{1}{12} \text{ of mass of one } {}^{12}\text{C} \text{ atoms}$$

$$= \frac{1}{12} \times 12 \times 6.023 \times 10^{23}$$

$$= 1.67 \times 10^{-24} \text{ gm}$$

$$= 1.67 \times 10^{-27} \text{ Kg}$$

Now,

$$m = 1 \text{ amu}$$

$$= 1.67 \times 10^{-27} \text{ kg}$$

Then,

$$E = mc^2$$

$$= 1.67 \times 10^{-27} \times (3 \times 10^8)^2$$

$$= 1.494 \times 10^{-10} \text{ J}$$

$$= \frac{1.494 \times 10^{-10}}{1.6 \times 10^{-19}} \text{ eV} [\because 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}]$$

$$= 931 \times 10^6 \text{ eV}$$

$$= 931 \text{ MeV}$$

$$= 931 \text{ MeV}$$

3) Explain why the mass of a nucleus is always less than the combined masses of its constituent particles.

Ans. We know that the mass of nucleus is always less than the combined mass of its constituent particles. This difference in mass is called mass defect. According to Einstein mass-energy relation, this mass is equivalent to an energy of $E = \Delta mc^2$ which is responsible for binding the nucleons together so as to form a nucleus.

Therefore, the mass defect is used up in binding energy for nucleons. Hence, the mass of a nucleus is slightly less than the combined masses of its constituents particle.

4) Why does a mountain of uranium not explode as a bomb?

Ans A mountain of uranium does not explode as a bomb because:

- i) It lacks critical mass
- ii) The concentration of fissile U-235 is too low (less than 0.7%).
- iii) Neutrons produced by spontaneous fission are absorbed by U-238 or escape before they can trigger a sustained chain reaction.

5) Why are neutrons used to initiate fission reaction?

Ans Neutrons being neutral particles can penetrate deep into the nucleus and provides the necessary excitation energy to split the nucleus.

6) Why is a neutron considered the most effective bombarding particle in nuclear reactions?

Ans Neutron is a neutral particle so it can hit the target nucleus without being repelled and is easily captured by the nucleus. It increases neutron to proton ratio inside the nucleus and makes the nucleus unstable. In this way it initiates nuclear reaction. On the other hand, being positively charged particles, if

protons and α -particles are used as bombarding particles they are repelled by the nucleus. Due to this reason, neutrons are most effective bombarding particles.

7) Which bomb is more explosive, hydrogen bomb or atom bomb? Explain.

Ans Hydrogen bomb is significantly more powerful. While an atom bomb relies on splitting heavy nuclei, a hydrogen bomb uses an atom bomb just as a 'trigger' to fuse light nuclei. Fusion releases much more energy per unit mass and has no limit on the amount of fuel that can be used.

8) Explain the significance of Einstein's mass energy relation.

Ans The significance lies in the law of Mass-Energy Equivalence. It proves that mass is a concentrated form of energy. It explains the massive energy released in Nuclear reactors, where a tiny decrease in mass results in a massive energy output.

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